Simulating Adaptive Behaviors in a Agent based Ecosystem: A Trait-Driven Grid Model

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Abstract:
Understanding the mechanisms that drive evolution and adaptation in complex biological systems remains a central challenge in computational biology. In this project, we present a flexible, open-source simulation software platform that models the evolutionary dynamics of interacting organisms on a two-dimensional grid. Each organism, represented as an agent with a set of mutable traits—including speed, memory, fear, energy efficiency, and lifespan—navigates a shared environment where food resources and predatory threats shape their survival and reproductive success.The simulation framework enables users to configure key ecological parameters, such as the number of herbivores, carnivores, and food sources, as well as trait mutation rates and environmental conditions. Organisms exhibit adaptive behaviors: herbivores forage for food and communicate predator threats, while carnivores pursue prey and may engage in cannibalism. Trait inheritance and mutation during reproduction allow for the emergence and selection of advantageous characteristics over time.Comprehensive visualization tools track population dynamics, trait distributions, and behavioral metrics across generations. By analyzing trends in memory, fear, and lifespan, users can investigate how individual and collective behaviors influence evolutionary outcomes. The model supports exploration of classic ecological questions, such as the optimal predator-prey ratio for stability, the evolutionary trade-offs between speed and resource efficiency, and the impact of social communication on group survival.This platform is designed to be modular and extensible, serving both as a research tool for investigating evolutionary hypotheses and as an educational resource for illustrating fundamental principles of adaptation and natural selection. By providing a dynamic, interactive environment for in silico experimentation, our project offers new opportunities to probe the interplay between genetics, behavior, and ecosystem structure in the evolution of complex life.

Keywords: Cellular Automaton,Emergent Behavior,Predator-Prey Dynamics,Evolutionary Simulation,Trait Inheritance,Grid-based Modeling

1. INTRODUCTION

1.1 The Symphony of Simple Rules

From the crystalline logic of von Neumann's self-replicating automata (von Neumann, 1940)—whose 29-state cellular machines first demonstrated mechanical reproduction—to Conway's pulsating Game of Life (Conway, 1970) with its gliders, pulsars, and infinite growth patterns, cellular automata (CA) have long revealed a profound truth: complexity blooms from simplicity. These grid-based universes, governed by elementary rules ($S_{t+1} = f(N(S_t))$ where N defines neighborhood states), mirror nature's own penchant for weaving intricate tapestries from basic threads.

This truth resonates through biological scales: in DNA's evolutionary dance modeled by Sirakoulis' base-pair automata (Sirakoulis et al., 2003), where

$$\text{Transition rules } \Psi : \{A, T, C, G\}^3 \to \{A, T, C, G\}$$

approximate mutation dynamics, and in ant colonies which were modelled in the book an introduction to genetic algorithms by (Mitchell, 1998) whose pheromone-based computation emerges from individual insect interactions. Yet classical CA, like Langton's self-replicating loops (Langton, 1986), remained frozen in a pre-Darwinian era—their static rule sets lacked the genetic variation ($\mu > 0$) and selection pressures ($\phi \propto P_{\text{pred}}$) that fuel biological innovation.

Our model rekindles von Neumann's original vision, marrying his kinematic self-reproduction blueprint with Hamilton's evolutionary game theory (Hamilton, 1964). Where traditional CA cells blindly follow transition tables, our agents evolve strategies through mutable traits:

$$\overrightarrow{g} = (v, \phi, m, \tau_m) \quad \text{where } \Delta g_i \sim \mathcal{N}(0, \sigma_g^2)$$

transforming Conway's deterministic playground into an adaptive ecosystem where survival demands genetic ingenuity.

1.2 Objectives

This study pursues three interlocked scientific goals through computational modeling:

- **Evolutionary CA Design**: Develop a cellular automaton framework that synthesizes von Neumann's self-replication (von Neumann and Burks, 1966) with Darwinian selection mechanisms, enabling the emergence of adaptive behaviors through trait mutation ($\Delta g_i \sim \mathcal{N}(0, \sigma_g^2)$) and environmental pressures
- **Trait Dynamics Analysis**: Characterize how interacting parameters—speed (v), fear (ϕ), memory (m)—govern organism fitness via mathematical equations to govern each of them
- **Complexity Bridge**: Demonstrate how local rules (r_v vision, r_c communication) generate global ecological patterns, connecting:
  - Microevolutionary processes (μ mutations/generation)
  - Population-level emergent phenomena (herd formation, predator-prey cycles)
- **Educational Synthesis**: Create an open-source platform for visualizing evolutionary trade-offs between:

$$\text{Fitness } \Phi = \alpha s - \beta f + \gamma m$$

  where s=speed, f=fear, m=memory, and α, β, γ are selection coefficients

Our work ultimately seeks to establish a new paradigm for studying evolutionary dynamics through programmable cellular ecosystems, bridging Wolfram's CA classifications (Wolfram, 2002) with modern population genetics (Hartl and Clark, 2022).

1.3 A New Evolutionary Overture

Our work reimagines this legacy through the lens of evolutionary theory, composing a digital ecosystem where predator and prey enact an adaptive ballet. Like von Neumann's universal constructor (von Neumann and Burks, 1966), agents self-replicate—but here, they also evolve. Each organism carries a genetic score of traits: *speed* (the allegro of escape), *memory* (a fading adagio of past encounters), and *fear* (a staccato rhythm modulating risk). These traits mutate under selection pressures, their frequencies swaying to the dual metronomes of predation and resource scarcity (Futuyma and Kirkpatrick, 2017).

The stage is a $n \times n$ grid—a microcosm where Moore neighborhoods dictate survival. Herbivores graze with Chebyshev efficiency, their movements a probabilistic fugue:

$$P_{\text{move}} = f_g \cdot (1 - \phi) + \epsilon_{\text{noise}}$$

where f_g is the food-seeking gene parameter and ϕ the fear coefficient. Carnivores stalk in counterpoint, their detection radius (r_d) scanning for prey. When predator meets herbivore, fear propagates through von Neumann neighborhoods (r_c), triggering synchronized retreats—a phenomenon mirroring real-world collective panic (Galef, 2012).

1.4 Bridging Scales, Unifying Theories

This CA framework harmonizes micro- and macroevolutionary scales. At the individual level, mutations perturb traits:

$$\Delta g_i \sim \mathcal{N}(0, \sigma_g^2), \quad \sigma_g \in [0.05, 0.15]$$

creating genetic drift melodies. Population-wide, these variations crescendo into emergent patterns: predator-prey oscillations echoing Lotka-Volterra dynamics, fear-induced herding akin to ecological synchronization (Schwabe and Wolf, 2012), and trophic cascades where trait adjustments ripple through the food web.

Six real-time visualizations—spanning spatial distributions, trait correlations, and lifespan trajectories—serve as our conductor's score, revealing how local rules orchestrate global complexity. The model extends Wolfram's CA taxonomy (Wolfram, 2002) into evolutionary time, demonstrating how Class IV behaviors (edge-of-chaos dynamics) foster sustainable ecosystems.

1.5 A Laboratory for Life's Algorithms

More than a simulation, this platform is a computational crucible. By tuning mutation rates ($\mu = 10^{-3}$/generation), resource distributions, and selection thresholds, users explore questions central to evolutionary biology:

- How does spatial memory trade off against metabolic cost?
- What fear levels optimize survival in fluctuating environments?
- When does cooperative herding outcompete solitary speed?

In the tradition of Lenia's continuous CA (Chan, 2023) and evolving neural CA (Mitchell, 1998), our work bridges theoretical biology and computational artistry. It offers both a research tool for probing evolutionary dynamics and a pedagogical sonata—a way to hear, in the interplay of simple rules, the grand symphony of life's adaptation.

2. METHODS

2.1 Agent-Based Model and Trait Space

We model an evolving community of organisms on a two-dimensional grid, where each agent is an individual with a vector of quantitative traits. The simulation extends classic cellular automata and agent-based models (Wolfram, 2002), but crucially, each agent's phenotype is determined by a set of evolvable, interdependent traits subject to explicit trade-offs.

2.2 Trait Definitions for Herbivores and Carnivores

The current model implements all the below traits with some basic rules for each of them.Some of these traits are unique to herbivores and carnivores

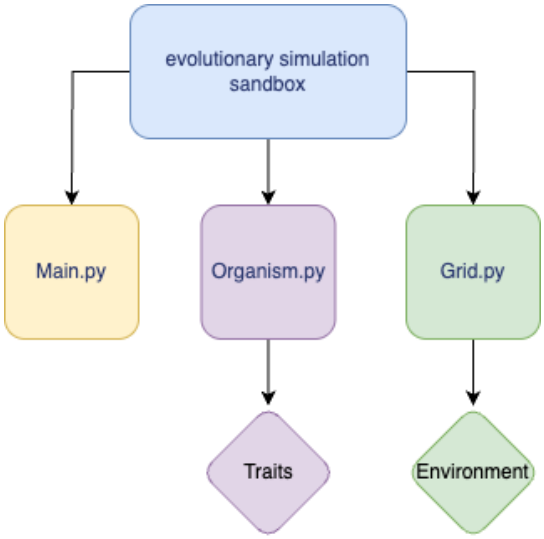


Fig. 1. Overall Methodology of the project

Table 1. Traits for herbivores and carnivores.

Trait	Herb.	Carn.
Lifespan (τ)	✓	✓
Speed (v)	✓	✓
Food Gene (θ_f)	✓	✓
Carnivore Detection (θ_c)	✓	✗
Stealth (σ)	✗	✓
Memory (m)	✓	✗
Energy Eff. (η)	✓	✓
Fear (ϕ)	✓	✗
Comm. Radius (r_{comm})	✓	✗
Carni. Sense (ψ)	✓	✗
Memory Cap. (C_m)	✓	✗
Hunt Strat. (S)	✗	✓

2.3 Trait Mutation at Reproduction

Upon division, each trait in the offspring is mutated independently using additive or multiplicative Gaussian noise, with values clamped to biologically plausible ranges. This approach follows established practices in agent-based evolutionary models (DeAngelis and Grimm, 2012; Grimm and Railsback, 2005; Szabó et al., 2022). The mutation strengths (standard deviations) are chosen to balance evolutionary exploration and trait stability, typically $\sigma \in [0.05, 0.5]$ for continuous traits (Szabó et al., 2022; Shaffer et al., 2017).

$$T' = \min(\max(T + \epsilon_T,\ T_{\min}),\ T_{\max}), \quad \epsilon_T \sim \mathcal{N}(0, \sigma_T^2) \tag{1}$$

where T is the parental trait, ϵ_T is the mutation, and $T_{\min}, T_{\max}$ are trait-specific bounds.

Specifically, in this model:

$$\theta' = \text{clip}(\theta + \epsilon_f,\ 0.05,\ 1), \quad \epsilon_f \sim \mathcal{N}(0, 0.5^2) \tag{2}$$

$$\text{v}' = \max(0.1, \text{v} + \epsilon_v), \quad \epsilon_v \sim \mathcal{N}(0, 0.1^2) \tag{3}$$

$$\eta' = \max(0.5,\ \eta\ e^{-|\epsilon_v|}), \quad \epsilon_v \sim \mathcal{N}(0, 0.1^2) \tag{4}$$

$$m' = \max(0, \text{memory} + \epsilon_m), \quad \epsilon_m \sim \mathcal{N}(0, 0.1^2) \tag{5}$$

$$\phi' = \max(0, \text{fear} - 0.5\epsilon_m) \tag{6}$$

$$d' = \max(1, \text{carnivore_detection} + \epsilon_d), \quad \epsilon_d \sim \mathcal{N}(0, 0.3^2) \tag{7}$$

$$L' = \max(100,\ L\ e^{-|\epsilon_d|/10}) \tag{8}$$

$$s' = \text{clip}(s + \epsilon_s,\ 0,\ 1), \quad \epsilon_s \sim \mathcal{N}(0, 0.05^2) \tag{9}$$

where $\text{clip}(x, a, b) = \min(\max(x, a), b)$.

For categorical traits (e.g., hunting strategy), mutation is implemented as random switching with fixed probability per division.

2.4 Movement and Foraging

Movement is governed by trait values and environmental cues, following stochastic behavioral rules (Grimm and Railsback, 2005; **?**):

- **Random movement:**
$$P(\text{move}) = \text{speed} \tag{10}$$
- **Directed foraging:** With probability food_gene, the organism moves toward the nearest food or a remembered food location.
- **Fleeing:** If a known carnivore is detected within detection radius d, the organism's fear increases and it attempts to move away:
$$\phi \leftarrow \min\left(1, \phi + \frac{0.8}{1 + \exp(-0.5(r - 3))}\right) \tag{11}$$
where r is the distance to the nearest threat. Escape is attempted with effective speed:
$$v_{\text{eff}} = \text{speed} \cdot \left(1 + 2.5 \cdot \phi^{0.7}\right) \tag{12}$$

2.5 Spatial Memory and Recall

Herbivores maintain a spatial memory of up to C food locations (here $C = 3$), each with a context and a decaying strength:

$$\text{strength}_i(t) = \frac{1}{(t - t_i + 1)^{\alpha}}, \quad \alpha = 0.5 + 0.3(1 - m) \tag{13}$$

Recall probability is context-dependent:

$$P(\text{recall location } i) \propto \text{strength}_i(t) \cdot \text{siml}(\text{context}_i, \text{current context}) \tag{14}$$

If memory is full, the weakest memory is replaced. This approach is supported by empirical and modeling studies on animal memory and spatial navigation (Fagan and et al., 2013; **?**).

2.6 Fear and Energy Feedbacks

Fear modulates other traits and physiological costs:

$$\eta \leftarrow \max(0.5, \eta - 0.1 \cdot \phi) \quad (15)$$

$$L \leftarrow L \cdot (1 - 0.3\phi) \quad (16)$$

$$\text{food_gene} \leftarrow \text{food_gene} \cdot (1 - 0.2\phi) \quad (17)$$

Such feedbacks are consistent with stress physiology and eco-evolutionary modeling (Creel and Christianson, 2014).

2.7 Communication

Herbivores communicate carnivore presence within a radius R:

$$R = \text{communication radius} \cdot (1 + 0.3\phi) \quad (18)$$

$$\phi_{\text{recv}} \leftarrow \min\left(1, \phi_{\text{recv}} + 0.4\left(1 - \frac{d}{R}\right)\right) \quad (19)$$

where d is the distance between sender and receiver. This mechanism is inspired by models of social learning and vigilance (Galef and Laland, 2022; **?**).

2.8 Aging and Death

At each time step, age is incremented by:

$$\boxed{\Delta\text{age} = \text{movement cost} \cdot (1 + \phi)} \quad (20)$$

An organism dies when age $\geq L$.

2.9 Reproduction

When conditions for reproduction are met, an offspring is created via the mutation process described above, inheriting the parent's traits with Gaussian noise and clamping.

Hyperparameter values: Mutation standard deviations for key traits are: $\sigma_f = 0.5$ (food_gene), $\sigma_v = 0.1$ (speed), $\sigma_m = 0.1$ (memory), $\sigma_d = 0.3$ (detection), $\sigma_L = 10$ (lifespan, as integer noise), $\sigma_s = 0.05$ (stealth). These values are in line with previous agent-based and eco-evolutionary modeling studies (Shaffer et al., 2017; Grimm and Railsback, 2005).

2.10 Emergent Eco-Evolutionary Phenomena

The interplay of softmax-constrained trait evolution, trade-off geometry, social learning, and environmental variability produces:

- **Herding and clustering** via local communication and spatial risk gradients
- **Trophic cascades** as predator-induced stress reduces herbivore foraging
- **Diversity maintenance** through dynamic niche partitioning and trade-off-driven coexistence

These outcomes are consistent with both theoretical and empirical findings in eco-evolutionary research

3. RESULTS

3.1 Biological Inference from Population Dynamics

Figure 2 illustrates coupled predator-prey dynamics under environmental variability. The herbivore population (blue) grows rapidly when resources are abundant, but shows sharp declines during drought-induced food shortages. Carnivore numbers (red, dotted) increase more gradually and are tightly linked to herbivore abundance, declining after prey crashes. These patterns highlight classic trophic cascades and demonstrate how bottom-up disturbances, such as drought, propagate through food webs and increase population volatility at multiple trophic levels. we noticed that in the equilibrium conditions when the system is not disturbed through environmental factors the ratio of herbivores to carnivores remain fixed

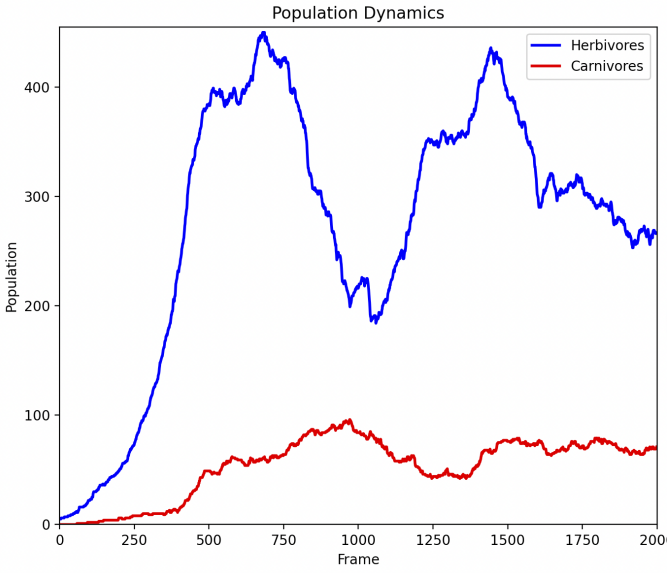


Fig. 2. Population Variation

3.2 Traits and their tradeoffs

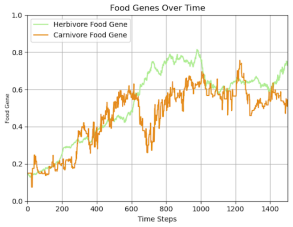
Evolution of Herbivore Memory and Lifespan Figure 3d shows the temporal dynamics of average memory (magenta) and lifespan (blue) in the herbivore population. Notably, lifespan declines sharply over time, dropping from initial high values to below 200 by the end of the simulation. In contrast, memory remains relatively stable, with only modest fluctuations around a mean value near 0.2. This decoupling suggests that while selection pressures or eco-evolutionary feedbacks are driving a reduction in lifespan, memory is maintained at a low but persistent level, possibly reflecting a minimum threshold required for effective foraging or predator avoidance.

Evolution of Food Gene in Herbivores and Carnivores Figure 3a tracks the mean food gene value—the propensity to move toward food—for both herbivores (green) and carnivores (orange). Both groups exhibit a clear evolutionary increase in this trait over time, with herbivores reaching higher values overall. The parallel rise in both populations suggests strong selection for efficient resource localization in both trophic levels, likely driven by competition and the spatial distribution of food. The convergence and fluctuations observed after time step 600 may indicate periods of resource scarcity or changing ecological interactions.

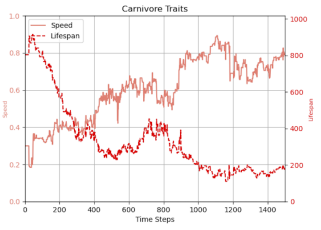
Evolution of Carnivore Speed and Lifespan Figure 3b presents the evolutionary trajectories of speed (solid red)

and lifespan (dashed red) in carnivores. There is a pronounced inverse relationship: as speed increases steadily throughout the simulation, lifespan declines sharply, stabilizing at low values. This pattern is consistent with a classic life-history trade-off, where increased investment in mobility or predatory efficiency comes at the cost of reduced longevity. The sustained high speed may reflect strong selection for effective prey capture in a competitive environment.

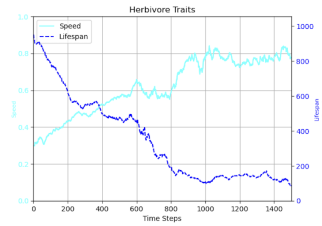
Evolution of Herbivore Speed and Lifespan Figure 3c displays the co-evolution of speed (solid cyan) and lifespan (dashed blue) in herbivores. Similar to carnivores, there is a marked trade-off: speed increases over time, while lifespan decreases. The rapid rise in speed suggests strong selection for enhanced escape or foraging ability, potentially in response to predation or resource competition. The concurrent decline in lifespan indicates that faster individuals may incur greater energetic or physiological costs, leading to reduced longevity. This result highlights the dynamic balance between survival strategies in the evolving population.



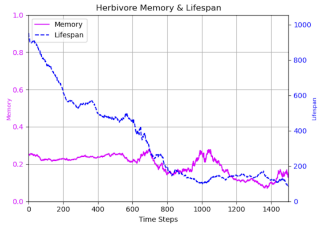
(a) Co evolution of food gene



(b) speed vs lifespan in carnivores



(c) speed vs lifespan in herbivores

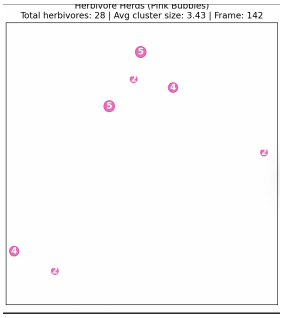


(d) memory and lifespan in herbivores

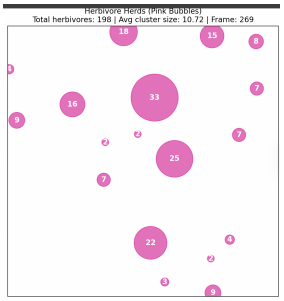
Fig. 3. Evolution of traits

3.3 Emergence of Herding

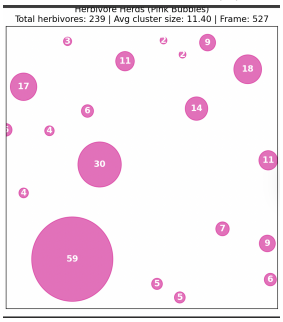
our model successfully demonstrates the spontaneous emergence of herding behavior among herbivores, as illustrated in Fig. 4. Without any explicit rule enforcing group formation, herbivores aggregate into distinct clusters on the lattice, a hallmark of collective movement and spatial self-organization in agent-based ecological systems (Alfonso Pérez-Escudero, 2013). This emergent herding arises purely from local interactions and trait-based decision-making, reflecting how simple behavioral rules and environmental cues can produce complex group-level patterns observed in nature. The ability of the model to reproduce such herding phenomena highlights its relevance for studying collective behavior and resource-use strategies in ecological communities.



(a) Intial Herding



(b) Intermediate Herding



(c) Herding at later stages

Fig. 4. Herding seen

3.4 Landscape of Fear and Herbivore Space Use

Our model vividly captured the emergence of a dynamic landscape of fear, where herbivores actively avoid areas of recent carnivore presence, even when food resources are available in those zones (see Fig. 5). The right panel shows a heatmap of "danger memory," with red regions indicating recent or recurrent carnivore activity. Herbivores (left panel, blue dots) are rarely observed in these high-risk (red) areas, instead clustering in blue (low-risk) zones. This spatial avoidance behavior is consistent with empirical and theoretical studies showing that prey modulate their foraging and movement to minimize predation risk, sometimes at the expense of foraging opportunities (Meredith S. Palmer, 2022; Clare McArthur, 2014). Such non-consumptive effects of predators-where risk alone, rather than direct predation, structures prey distributions-are a hallmark of the landscape of fear framework and highlight the complex interplay between food and fear in shaping ecological communities.

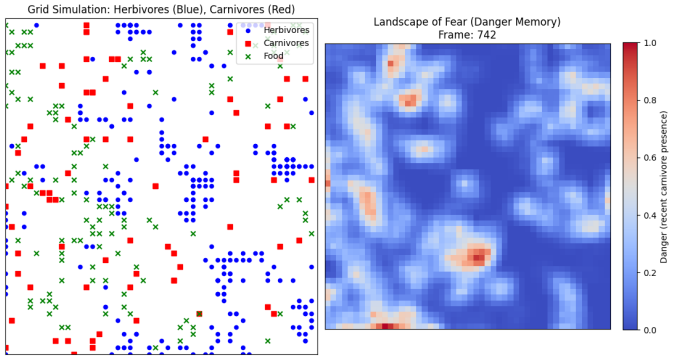


Fig. 5. Landscape of fear

4. DISCUSSION

This work presents a flexible, trait-based agent-based modeling (ABM) framework for investigating eco-evolutionary dynamics in complex ecosystems. Unlike classic models that rely on explicit trade-off constraints or softmax allocation mechanisms (Zakharova et al., 2019a; Shoval et al., 2012), our updated approach encodes each trait as an independent, continuous variable subject to direct mutation and biologically motivated feedbacks. This design enables the simulation of a wide range of trait combinations and evolutionary trajectories, while still capturing key eco-evolutionary processes such as spatial memory, social information transfer, and fear-modulated behavior.

A major advantage of this framework is its capacity to represent inter-individual variability and context-dependent decision-making, both of which are increasingly recognized as critical drivers of ecological and evolutionary dynamics (Donald L. DeAngelis, 2018; Chevalier et al., 2022; Zakharova et al., 2019a). By allowing traits to mutate independently (with clamping and feedbacks), the model can capture the emergence of behavioral syndromes, personality variation, and the evolution of context-sensitive strategies (Bolnick et al., 2011). The inclusion of spatial memory, communication, and stress/fear feedbacks enables the simulation of emergent phenomena such as herding, trophic cascades, and eco-evolutionary feedbacks, in line with recent theoretical and empirical studies (Chevalier et al., 2022; Hendry and et al., 2021; Zakharova et al., 2019a).

Model strengths and extensibility. The absence of rigid trade-off surfaces or allocation constraints makes the framework highly extensible: researchers can readily add new traits, interaction types, or environmental drivers (Zakharova et al., 2019a,b). This flexibility is particularly valuable for exploring the consequences of intraspecific trait variation, the evolution of behavioral plasticity, and the effects of rapid environmental change on community structure (Hendry and et al., 2021; Zakharova et al., 2019a). The agent-based structure also facilitates the integration of empirical trait data and the investigation of trait-mediated species interactions across scales (Zakharova et al., 2019a,b).

Limitations and challenges. However, this flexibility comes with important caveats. First, the lack of explicit trade-off constraints may allow for unrealistic trait combinations unless biologically justified feedbacks or bounds are imposed (Zakharova et al., 2019a; Donald L. DeAngelis, 2018). Without careful calibration, traits may drift toward extremes or exhibit correlations not observed in real systems. Second, the increased number of free parameters and mutation rates introduces challenges for parameterization, sensitivity analysis, and empirical validation (Zakharova et al., 2019b; Hendry and et al., 2021). Third, as with all ABMs, computational cost can become significant for large populations or long simulations (Techniques, 2020).

Future directions. To address these challenges, future work should focus on systematic parameter tuning, integration with empirical trait distributions, and the development of standardized protocols for model validation and comparison (Zakharova et al., 2019a,b). Incorporating more mechanistic or empirically derived feedbacks, such as metabolic trade-offs or resource constraints, may further enhance biological realism (Hendry and et al., 2021). Finally, community-driven development of open, modular model components will facilitate reproducibility and cross-system comparisons, advancing the field of trait-based eco-evolutionary modeling (Zakharova et al., 2019a; Chevalier et al., 2022).

In summary, our framework demonstrates the power and flexibility of direct, feedback-based trait mutation in ABMs, while highlighting the importance of careful model design, parameterization, and validation. As trait-based and agent-based approaches continue to mature, they will play an increasingly important role in advancing predictive, mechanistic understanding of biodiversity dynamics under global change.

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